

AI-Powered Document Upload & Task Extraction Report

Automated Requirement Ingestion, Intelligent Task Generation & Dual-Mode UI Implementation

1. Executive Summary

Today's development focused on implementing a robust, end-to-end **AI Document-to-Task Ingestion Engine** inside the Kanban Task Board application. Users can now drag and drop or browse project documents (PDF, Word DOCX, Markdown, Text, CSV), and the system autonomously parses the file content, passes it through Google Gemini 2.5 Flash with strict validation guardrails, and auto-populates all task fields including title, structured markdown description, assignees, priority, labels, and future due dates.

The feature was designed as a seamless dual-mode experience inside the **Create Task** modal, allowing users to toggle effortlessly between natural language prompt creation and rich document extraction.

2. Key Accomplishments & Deliverables

- **Multi-Format File Parser:** Built a unified document processor supporting native multimodal PDF ingestion via base64, Word documents (.docx) via Mammoth library, and plaintext/markdown (.txt, .md, .csv) with a strict 10MB size limit.
- **Gemini 2.5 Flash Multimodal Pipeline:** Configured Gemini 2.5 with structured JSON schema output, extracting actionable task parameters while filtering out unstructured or irrelevant text.
- **Strict Content & Relevance Guardrail:** Integrated an anti-hallucination filter that validates whether an uploaded file contains actionable project work, gracefully rejecting non-task inputs (e.g., songs, poetry, casual chat, recipes) with descriptive user feedback.
- **Intelligent Entity Matching:** Automated matching of assignees and labels by passing active board members and label registries to the prompt context.
- **Strict Future Due-Date Resolution:** Built deterministic date anchoring with day-of-week context, ensuring inferred deadlines are strictly on or after today's reference date and rejecting historical dates.
- **Interactive Drag & Drop UI:** Built an animated, accessible file uploader component with drag-and-drop feedback, file size display, loading states, error dismissals, and one-click form auto-fill.

3. End-to-End Technical Architecture

Step / Layer	Component File	Technical Responsibility & Operation
1. Client UI	document-task-uploader.tsx	Handles drag-and-drop, validates file type & 10MB limit, bundles FormData with assignee/label IDs and reference timestamp.
2. Server Action	src/lib/actions/ai.ts	Verifies session authentication, unpacks FormData, orchestrates document parsing, invokes Gemini API, validates output with Zod.
3. Doc Processor	src/lib/ai/document-parser.ts	Dispatches parser based on MIME/extension: Base64 for PDF, Mammoth for DOCX, UTF-8 buffer decoding for TXT/MD/CSV.
4. Prompt & Guardrail	document-task.prompt.ts	Constructs system instructions, relevance filter, date anchor, entity resolution schema, and structured JSON output rules.
5. AI Model	src/lib/ai/gemini.ts	Invokes Gemini 2.5 Flash with multimodal inline data or text content, enforcing deterministic JSON schema formatting.
6. Form Auto-fill	create-task-form.tsx	Applies extracted payload to form state: updates title, description, assignee, priority, labels, and due date with visual confirmation.

4. Detailed Implementation Breakdown

A. Document Parser Engine (src/lib/ai/document-parser.ts)

A dedicated server-only parser module that safely inspects and extracts text or binary representations from user uploads:

- **PDF Handling:** Converts raw buffer into Base64 inline data for Gemini's native multimodal vision/document parser, preserving complex tables and structural context without external PDF OCR dependencies.
- **Word Processing:** Utilizes mammoth to extract raw, semantic text from Word .docx archives while discarding proprietary XML bloat.
- **Text/Markdown/CSV:** Decodes UTF-8 buffers with minimum character validation checks to reject empty or corrupted files.
- **Security Constraints:** Strictly checks 10MB size limit and throws strongly-typed AppError on malformed payloads.

B. AI Prompt Engineering & Content Guardrails (src/lib/ai/prompts/document-task.prompt.ts)

Designed comprehensive system prompts and JSON schemas with strict behavioral constraints:

- **Relevance & Actionability Filter:** Introduces isValidTaskDocument and rejectionReason. If the document contains recipes, songs, poetry, or noise, the AI rejects it with an explanatory explanation instead of hallucinating tasks.
- **Temporal Anchoring:** Injects dynamic date anchoring (e.g. 'Sunday, 2026-03-01') so relative dates like 'next Wednesday' or 'by end of quarter' are resolved accurately to future ISO dates.
- **Entity Resolution:** Compares document text with existing team members and label catalogs, auto-mapping valid internal UUIDs.
- **Structured Description Formatting:** Synthesizes document findings into clean Markdown sections: Overview, Deliverables, and Acceptance Criteria.

C. Interactive Document Upload Component (src/components/ai/document-task-uploader.tsx)

A modern, reactive UI component adhering to design standards:

- **Drag & Drop Zone:** Visual dashed border with active drag highlight, file icon, size indicator, and remove/clear buttons.
- **Asynchronous Status States:** Includes idle state, loading state with spinning loader ('Extracting...'), dismissable error banners, and success feedback.
- **Safe Client-Side Validation:** Checks file extension and size prior to network dispatch to avoid wasted API calls.

D. Modal Form Integration (src/components/board/create-task-form.tsx)

Integrated the uploader directly into the Kanban creation flow:

- **Dual AI Modes:** Added an intuitive segmented toggle allowing users to choose between **Prompt Mode** (Magic Task Input) and **Document Mode** (File Upload).
- **Instant Field Population:** Upon extraction, fields (Title, Description, Assignee, Priority, Labels, Due Date) populate automatically, allowing the user to review and tweak before final submission.

5. Format Compatibility & Guardrail Matrix

File Format	Extension	Processing Method	Max Size	Guardrail Status
PDF Documents	.pdf	Native Multimodal Base64	10 MB	Active & Verified
Word Documents	.docx	Mammoth Raw Text Extraction	10 MB	Active & Verified
Markdown Specs	.md	UTF-8 Direct Buffer Parse	10 MB	Active & Verified
Plain Text Files	.txt	UTF-8 Direct Buffer Parse	10 MB	Active & Verified
CSV Data / Notes	.csv	UTF-8 Direct Buffer Parse	10 MB	Active & Verified
Non-Task Files	Any	Relevance Filter Analysis	N/A	Graceful Rejection

6. Quality Assurance & Edge Case Handling

- **Empty or Corrupted Files:** Checked via buffer length thresholds; throws clear user error.
- **Historical Dates in Document:** Guardrail ensures any date prior to today's date is dismissed (returns null), preventing accidental past-due tasks.
- **No Assignee / Label Match:** Gracefully leaves fields blank for manual selection without breaking the creation flow.
- **Large Files (>10MB):** Instantly caught on client-side before upload to conserve bandwidth and server memory.

7. Summary & Future Roadmap

The Document Upload & Task Extraction feature is fully implemented, verified, and integrated into the Kanban board. It significantly accelerates project management workflows by turning complex technical briefs into organized, actionable tasks in seconds. Future roadmap enhancements may include multi-task extraction from single documents (batch splitting) and PDF visual attachment retention.

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